

Marking Guide for Advanced Software Engineering Course Projects

Reference: Craig Larman, *Applying UML and Patterns: An Introduction to Object-Oriented Analysis and Design and Iterative Development* (3rd Edition)

Methodology: Unified Process (UP) – Iterative & Incremental (Not Waterfall)

Duration: 15 Weeks

This marking guide follows UP phases (Inception, Elaboration, Construction, Transition) and iterative development, as emphasized in Larman's Point-of-Sale (POS) case study.

Project Breakdown & Milestones (15 Weeks – UP Phases)

Total Marks: 100

Week	UP Phase	Milestone	Deliverables	Marks
Week 1-2	Inception	Vision & Feasibility	- Problem statement & business case - High-level requirements (10% use cases) - Risk list & feasibility study - Iteration plan (UP phases mapped)	10
Week 3-5	Elaboration (Iteration 1)	Core Architecture & Requirements	- 30% use cases detailed (key scenarios) - Domain model (conceptual classes) - System sequence diagrams (SSDs) - Architectural proof-of-concept (e.g., layered design)	20
Week 6-8	Elaboration (Iteration 2)	Risk-Driven Design	- Remaining use cases (70%) - Design class diagrams (refined) - UI prototypes (if applicable) - Updated risk list & mitigation	15

Week	UP Phase	Milestone	Deliverables	Marks
Week 9-12	Construction (Iterations 3-4)	Feature Implementation	- Prioritized features implemented - Unit & integration tests - Continuous integration (Git) - Defect tracking	30
Week 13-14	Transition (Iteration 5)	Polishing & Deployment Prep	- Beta testing & user feedback - Performance tuning - Final documentation (user manual, models)	15
Week 15	Transition	Demo & Final Submission	- Live demo (working system) - Final report (lessons learned, UP reflection) - Code & documentation repository	10

Detailed UP-Aligned Marking Criteria

1. Inception (10 Marks)

- **Vision document** (business objectives, scope) – 3
- **10% use cases (high-level)** – 3
- **Risk list & iteration plan** (UP phases justified) – 4

2. Elaboration – Iteration 1 (20 Marks)

- **30% use cases (detailed, with SSDs)** – 6
- **Domain model (UML class diagram)** – 5
- **Architectural proof-of-concept** – 5
- **Updated risk list** – 4

3. Elaboration – Iteration 2 (15 Marks)

- **Remaining use cases (70%)** – 5

- Design class diagrams (refined with methods) – 5
- UI prototype (if applicable) – 3
- Risk mitigation progress – 2

4. Construction – Iterations 3-4 (30 Marks)

- Prioritized features implemented (per iteration plan) – 10
- Test cases (unit + integration) – 10
- Version control & CI adherence – 5
- Defect tracking log – 5

5. Transition – Iteration 5 (15 Marks)

- Beta testing report (user feedback) – 5
- Performance/security fixes – 5
- Final documentation (updated models, user manual) – 5

6. Demo & Final Submission (10 Marks)

- Live demo (key scenarios shown) – 5
- Final report (UP process reflection, lessons learned) – 3
- Code repository (Git) & completeness – 2

Key UP Principles Emphasized (Per Larman's POS Example)

1. **Risk-Driven:** Early iterations address high-risk areas (e.g., architecture).
2. **Use-Case Driven:** Requirements captured as use cases, refined iteratively.
3. **Architecture-Centric:** Core design validated in Elaboration.
4. **Iterative & Incremental:** Features delivered in prioritized chunks.

Penalties & Bonuses

- **Late submissions:** -10% per week.
- **Plagiarism:** Zero tolerance.
- **Bonus (up to 5%):** Advanced patterns (e.g., GoF), CI/CD automation, or exceptional UX.

Example: POS System Alignment

- **Inception:** "Discover core requirements for a POS system (e.g., process sale, handle returns)."
- **Elaboration:** "Model Sale, Payment, Product classes; design layered architecture."
- **Construction:** "Implement processSale() with payment validation."
- **Transition:** "Optimize database queries; prepare installation guide."